College Management System Database Setup Guide

Database: PostgreSQL | **Tool:** pgAdmin 4 | **Type:** Database-only project (no application front end)

This guide shows you, step by step, how to set up the College Management System database on a new computer, load the data from the CSV files, check that everything is correct, and run the analysis queries. You do not need prior SQL knowledge. Follow the steps in the order given.

Setup Guide

Run Order: What to Run First

Step	What to do	File / action
1	Create an empty database named college_db	pgAdmin: right-click Databases, Create, Database...
2	Create the 8 empty tables (plus trigger, procedure, views)	01_schema_only.sql
3	Import the 8 CSV files in the exact order shown below	pgAdmin: right-click table, Import/Export Data
4	Fix the auto-ID counters	02_fix_sequences.sql
5	Check that the import is correct	03_verify_import.sql
6	View every table, one by one	view_all_tables.sql
7	Run the 229 analysis queries, one at a time	complete_data_analysis.sql
Only if an import fails	Delete all rows and start the import again from step 3	99_DANGER_reset_data.sql

CSV Import Order (Step 3)

Import the files strictly in this order. Parent tables must be loaded before the tables that depend on them (foreign keys).

Order	Table	CSV file	Needs first
1	department	01_department.csv	none
2	student	02_student.csv	department
3	faculty	03_faculty.csv	department
4	course	04_course.csv	department, faculty
5	exam	05_exam.csv	course
6	result	06_result.csv	exam, student
7	enrollment	07_enrollment.csv	student, course
8	attendance	08_attendance.csv	enrollment

How to Import One CSV File in pgAdmin

- Expand college, then Tables. Right-click the table for the current step and choose Import/Export Data...
- General tab: switch to Import, choose the CSV file, Format = csv, Encoding = UTF8.
- Options tab: Header = ON, Delimiter = , (comma). Leave NULL Strings empty.
- Columns tab: keep all columns selected (the ID column must be included).
- Click OK. The process message must say Successfully completed. Then go to the next table.

How to Run a Script or a Query

- Right-click college_db, choose Query Tool, then open the .sql file with the folder icon.
- Setup scripts (01, 02, 03): nothing highlighted, press F5 once.
- Analysis files: highlight ONE query, then press F5. pgAdmin shows only the last result when many queries run together.

- Blocks that start with BEGIN; and end with ROLLBACK; must be highlighted completely and run together.

Do and Do Not

Do	Do not
Run 01, then the CSV import, then 02, then 03, in this order.	Do not import a CSV file twice.
Run each setup script only once.	Do not import into a table before its parent tables are loaded.
Run analysis queries one at a time (highlight, F5).	Do not open the CSV files in Excel and save them (it breaks dates).
Run BEGIN; ... ROLLBACK; blocks as a whole.	Do not run only the DELETE or TRUNCATE line of a BEGIN; ... ROLLBACK; block.
Use 99_DANGER_reset_data.sql only if an import failed.	Do not run the whole analysis file in one go.

Which Analysis Queries Are Safe (complete_data_analysis.sql)

Queries	What they do	Notes
Q1 - Q165	Read-only analysis	Safe. Any order, one at a time.
Q166 - Q171	Views	Run in order (Q167 before Q171, Q169 before Q170).
Q172 - Q186	Indexes, query plans, database information	Safe.
Q187 - Q201	Test ALTER, INSERT, UPDATE, DELETE, TRUNCATE	Only as a complete BEGIN; ... ROLLBACK; block.
Q202 - Q213	Functions, procedure, audit table, triggers	Optional. Permanent additions. Run once. Q207 and Q212 are BEGIN/ROLLBACK tests.
Q214	Test of the no-future-attendance rule	Commented out. When enabled it must fail with an error (that is correct).
Q215 - Q229	Lists, roles, data quality checks, JSON, dashboard	Q217 - Q220 are optional and need the postgres user. The rest are safe.